Small Area Estimation with the Fay-Herriot Model

ICAR and Logit Transformation

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1. Introduction

1.1. Background

Much of the data that statisticians rely on for their work is taken for granted.

1.2. Methods

2. The Data

2.1. Data Sources

2.1.1. 2020 Decennial Census of Population and Housing

The first source of data for this project comes from the 2020 Decennial Census. Mandated by the Constitution of the United States, the census is carried out every 10 years by the U.S. Census Bureau and "determines the number of people living in the United States" [1]. Though the primary purpose of the census is to distribute seats in the U.S. House of Representatives [1], it is also the official source of high level demographic and housing data for the entire country. As such, it is used as the source of truth in this project. Every single model will be compared to the values provided by the 2020 Decennial Census to evaluate the quality of predictions. It is important to note that, though treated as the source of truth, that the values provided by the U.S. census are not perfect. An analysis performed by the U.S. Government Accountability Office in 2024 found 14 states exhibiting statistically significant net coverage errors in the 2020 census, though Iowa is absent from this list [2]. Accounting for this uncertainty is outside the scope of this project, so all values provided by the census are treated as the true values.

2.1.2. American Community Survey

The second, and primary source of data for this project comes from the American Community Survey (ACS). The ACS is an ongoing survey performed by the U.S. Census Bureau about the population of the United States of America [3]. The Census Bureau is primarily known for the U.S. Decennial Census, but the lesser known ACS serves as a crucial companion piece to it. Instead of every 10 years, the ACS is sent out to a sample of addresses every single month of every single year [4]. The ACS in this aspect fills a very important role, as it is able to provide current information between decennial census collections, and is also able to cover a more robust set of topics than the census. The tradeoff is that the ACS comes from a sample of households, so every result is met with a certain degree of uncertainty. However, unlike traditional surveys, individuals selected for the ACS are legally obligated to respond to it [5]. The consequence of this is that the uncertainty in the ACS results can be treated as a known quantity, which allows for mathematical techniques that are not normally possible with traditional surveys that are optional to respond to.

The data from the ACS comes in two primary forms, 1-year and 5-year estimates. The 1-year estimates are less precise but more current, encompassing 12 months of collected data but only providing information on areas with a population size of at least $65,000$ [6]. The

5-year estimates are based on 60 years of collected data, and so are more precise but less current. The 5-year estimates also provide information on all areas surveyed, so this even includes extremely sparsely populated county or census tract level data [6].

This project will be using the 2020 5-year estimates, encompassing survey data from 2016-2020 for each county in Iowa. The variables used are `B25002_001` and `B25002_003` which correspond to the number of total housing units and vacant housing units respectively. The ratio of the two is used to calculate a vacancy rate for each county in Iowa which functions as the response for this project.

2.1.3. NASS Crop Data Layer

The third source of data for this project is known as the Crop Data Layer, or CDL. According to the USGS, the CDL, created by the USDA National Agricultural Statistics Service (NASS), is "an annual raster, geo-referenced, crop-specific land cover data layer produced using satellite imagery and extensive agricultural ground reference data" [7]. The USDA website carries direct downloads to every annual data layer going back to 2008. The year used for this analysis is the same as the decennial census, 2020. This dataset contains per-pixel cropland classifications at a 30-meter resolution for the entire country. The classifications were done using a machine learning model trained primarily on the Farm Service Agency (FSA) Common Land Unit (CLU) data, though some states incorporate additional auxiliary data from other agencies and institutions [7]. This data within the region of Iowa is used to create the spatial covariate for the Fay-Herriot model. Incorporation of pixel classification uncertainty in the CDL is outside the scope of this project, so all classifications are treated as the true classifications.

2.2. Data Processing

2.2.1. American Community Survey

ACS data was acquired for this project through the Census Bureau API, indirectly interfaced with via the `tidycensus` R package on CRAN. Data was pulled from the 2020 5-year ACS estimates for the `B25002_001` and `B25002_003` variables, which as previously mentioned correspond to the total housing units and vacant housing units per county. The resulting dataframe has a grain of one row per variable per county, so 2 rows per county. Each row contains the estimate for that variable, its corresponding Margin of Error (MoE) and the spatial geometry of that county. This dataframe was pivoted to a grain of one row per county to facilitate creation of the response variable, vacancy rates. For each county, the vacant unit count was divided by the total housing unit count to get the vacancy rate of that county.

$$\text{Vacancy Rate} = \left(\frac{\text{Vacant Housing Units}}{\text{Total Housing Units}} \right)$$

To get the standard deviation of this new estimate, two steps are necessary. First, a conversion from MoE to standard error is required for each estimate. Then, the standard errors of both estimates must be combined to form the standard error of the proportion. The Census Bureau

provides guidelines on both steps, and the same document will be referred to for both [8]. Conversion from MoE to SE is done by simplying dividing the MoE by 1.645.

$$\text{Standard Error} = \frac{\text{Margin of Error}}{1.645}$$

As for the standard error of the proportion response, the following setup is used. First, let $P = \frac{A}{B}$, where A is a subset of B. A will represent the vacant housing units, B will be represent all housing units. Then, the standard error of a counties housing vacancy rate can be represented as

$$\text{SE}(P) = \frac{1}{B}\sqrt{\text{SE}(A)^2 - P^2 \times \text{SE}(B)^2}$$

2.2.1.1. Logit Transformation of the Response

As the Fay-Herriot model relies on linear regression as a key component, a proportion response can behave poorly. The domain of the output of a linear reression is the entire real line, whereas a proportion is bounded between 0 and 1. In a more traditional scenario this would lead to a logistic regression model, though usage of the Fay-Herriot model requires alternative methods to overcome this limitation. Logit transformations have been used in this scenario before, such as in research performed by Cucu Sumarni et al. that found a "logit transformation within the SAE framework improved the precision of proportion estimates" [9]. This takes a response bounded between 0 and 1 and maps it to the entire real line. Thus, our transformed response can be represented as

$$Q = \ln\left(\frac{P}{1-P}\right)$$

2.2.1.2. Logit Transformation of the Standard Error

Yet again, additional work is required for the conversion of the standard error. For this, the delta method was utilized. For reference,

Definition 2.2.1.2.1 (Delta Method)**.** Let Y_n be a sequence of random variables that satisfies $\sqrt{n}(Y_n - \theta) \to N(0, \sigma^2)$ in distribution. For a given function g and a specific value of θ, suppose that $g'(\theta)$ exists and is nonzero. Then

$$\sqrt{n}[g(Y_n) - g(\theta)] \to N\left(0, \sigma^2[g'(\theta)]^2\right)$$

in distribution.

To apply this, let

$$g(P) = \ln\left(\frac{P}{1-P}\right)$$

Then,

$$g'(P) = \frac{1}{P - P^2}$$

We can then substitute this into the definition to get the updated variance, σ_Q^2.

$$\begin{aligned}\sigma_Q^2 &= \sigma^2[g'(P)]^2) \\ &= \sigma^2\left(\frac{1}{P-P^2}\right)^2 \\ &= \frac{\sigma^2}{(P-P^2)^2} \\ &= \frac{\sigma^2}{P^2(1-P)^2}\end{aligned}$$

From this, our transformed standard error is

$$\sigma_Q = \frac{\sigma}{P(1-P)}$$

The final dataframe contains a variety of other columns for diagnostic and exploratory purposes, but the dataframe of modeling components looks like so:

County	Vacancy Rate	SE	Vacancy Rate (Logit)	SE (Logit)	Geometry
Polk	0.064	0.003	−2.69	0.05	-
Grundy	0.076	0.011	−2.50	0.154	-
-	-	-	-	-	-

Table 1: Snippet of ACS Dataframe used for modeling. Untransformed columns and geometry included for the sake of reference and demonstration.

2.2.2. 2020 Decennial Census of Population and Housing

Ingestion and processing of the 2020 decennial census data was handled in a similar manner to the ACS data. Data was acquired through the `tidycensus` package, used to access the 2020 Census Demographic and Housing Characteristics File (DHC). The variables pulled were `H3_001N` and `H3_003N`, which correspond to total housing units and total vacant units repsectively. The grain of the data is the same as the original ACS data, one row per county per variable. The final table had one row per county, and a new column representing the housing vacancy rate was created in percent units. These values were not passed through a LOGIT transformation as they are intended for comparison against the predictions after they have been transformed back into the original proportion scale.

To ensure the accuracy of census housing vacancy rates, output was compared to the official housing vacancy numbers provided by the Census Bureau in their 2020 Iowa breakdown [10]. All of the values matched their values presented on that sites dashboard. Below is a snippet of the data.

County	Total Housing Units	Vacant Housing Units	Vacancy Rate
Polk	210184	13293	0.06
Grundy	5465	396	0.07
-	-	-	-

Table 2: Snippet of the Census DataFrame containing the true values for model evaluation. Non-proportion columns retained for the sake of reference and demonstration.

2.2.3. NASS Crop Data Layer

The crop data layer requires careful processing to be usable in this project. At a 30×30 meter resolution, the 2020 crop data layer raster image has a rectangular dimension of 96523×153811 pixels, or 14.8 billion pixels total. About 6.2 billion (42%) pixels are empty background pixels and 8.7 billion encompassing actual categorizations. Naive calculations performed on this object are prone to overloading the memory of the machine running them. The `terra` package was used for loading and processing this file. For filtering down to only Iowa's portion of the CDL, the `tigris` package was utilized to download Iowa's TIGER shapefiles from the U.S. Census. The coordinate reference systems (CRS) of the two objects were aligned to NAD83 Conus Albers (ESPG:5070). Cropping and masking is then performed on the CDL using the Iowa shapefile object. Cropping filters the CDL down to the rectangular region containing Iowa, and masking changes all pixels outside of the state borders to NA.

This new object, the Iowa CDL, contains a rectangular dimension of 11671×17794, or approximately 207 million pixels total. That is 1.3% the size of the original CDL. Of those, about 46 million (22%) are empty background, and appoximately 162 million (78%) pixels are categorized. From here, the goal is to get a table containing the frequency and proportion breakdown of every county in the state. To accomplish this goal, a similar process was used as before. County shapefiles were acquired using `tigris` and, using an iterative approach over the counties that leverages cropping and masking, the frequencies of every CDL classification were captured alongside the total number of pixels within the county's borders. A dataframe was created from this that contains one row per county, with the percent of the pixels in the county that were classified as either corn or soybeans. The proportion from this functions as the spatial covariate for the model and can be represented as:

$$\text{Proportion Corn or Soy}_i = \frac{\text{\# of Corn Pixels}_i + \text{\# of Soybean Pixels}_i}{\text{\# of Pixels Total}_i}$$

With i representing the index of a given county.

Below is a snippet of the resulting dataframe. Values are saved as percentages and converted to proportions at the modeling stage.

County	Soybeans (%)	Corn (%)	Corn or Soy (%)
Polk	17.8	22.4	40.2
Grundy	36.0	52.5	88.5
-	-	-	-

Table 3: Snippet of the CDL DataFrame containing the spatial covariate of the model. The individual soy and corn columns are kept for reference.

2.3. Data Analysis

2.3.1. Iowa Spatial Structure

Iowa was chosen for this project for a few reasons. The first is that it has some very populated counties and very sparsely populated counties. This allows for a good demonstration of the

Fay-Herriot model as it can effectively pool from informative counties. This will be discussed in more detail in later sections.

Second is the layout of its counties. Iowa has 99 counties total and they are arranged in a very neat grid as shown in the figure below. On average, each county has 5.9 neighbors with a minimum of 3 and a maximum of 8. There are no disjointed islands or bizarre geometries that would complicate spatial calculations as well. As such, this state shows useful properties for both Fay-Herriot modeling and spatial methods.

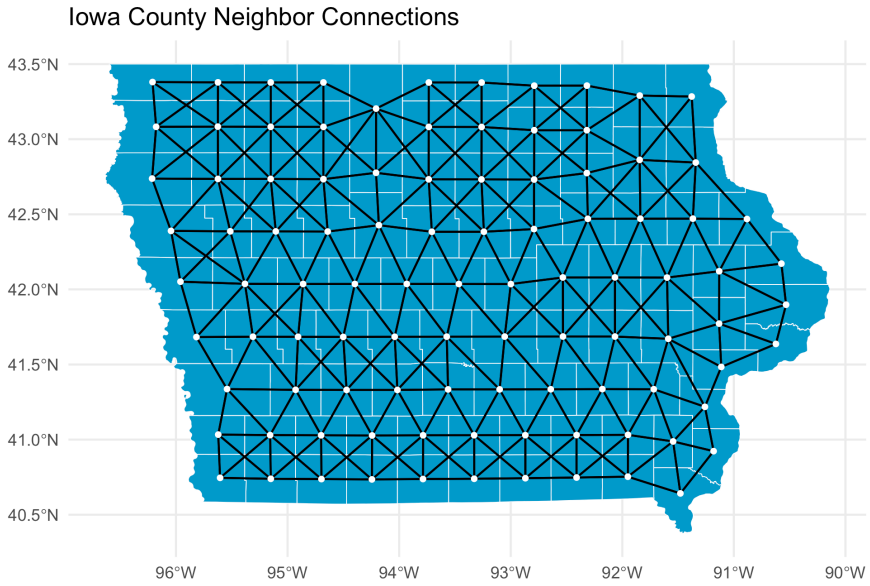


Figure 1: Plot showing adjacent county links across Iowa.

2.3.2. 2020 Decennial Census of Population and Housing

The counties in Iowa vary dramatically in terms of their population and housing. The number of total housing units can range from as little as 1,888 (Adams County) all the way up to 210,184 (Polk County). 29 of the 99 total counties have less than 5,000 total housing units.

Metric	Total Units	Vacant Units	Vacancy Rate
Minimum	1888	274	0.05950
1st Quartile	4684	518	0.07892
Median	7037	701	0.10144
Mean	14271	1255	0.10842
3rd Quartile	10947	1131	0.11803
Maximum	210184	13293	0.41743

Table 4: Summary of the housing data from the 2020 decennial census. (N=99)

The same trend is seen with vacant units, as one would expect places with more houses also have more vacant houses. These variables show an almost perfect correlation ($r = 0.947$). This relationship does not hold however for vacancy rates which show a mild negative correlation with total housing units ($r = -0.237$). The highest vacancy rates appear on counties

with low total housing units. This is unsurprising as each vacant house contributes more to the vacancy rate than one would in larger counties.

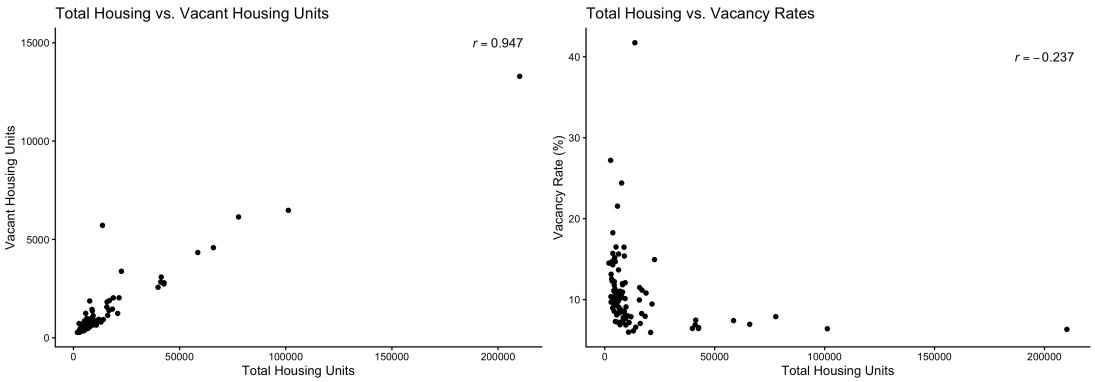


Figure 2: Plots showing the change of relationship when moving from pure vacant housing units to a vacancy rate.

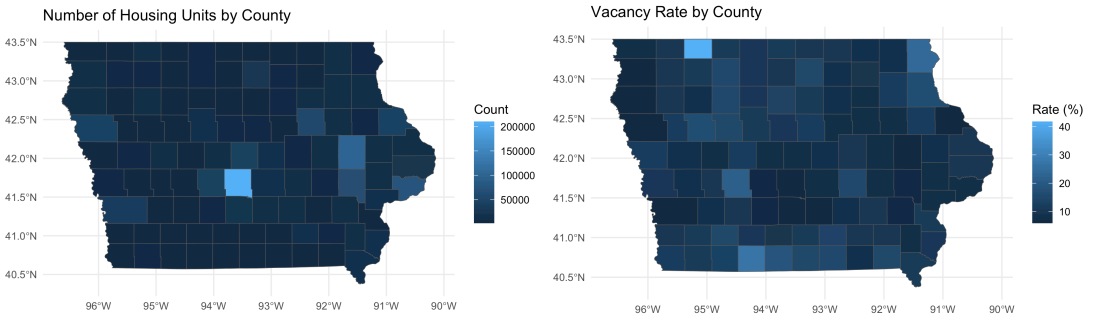


Figure 3: Spatial plots showing the distribution of housing units and vacancy rates by county across the state.

Examining these values at the spatial level, we can see that the overall population of housing units is heavily concentrated in a small number of counties scattered throughout the state. In contract, vacancy rates show a far more mixed distribution of values. Visually it is unclear if there is evidence of spatial autocorrelation for the vacancy rates, but there does appear to be some grouping.

2.3.3. American Community Survey

To begin the analysis, we examine the ACS equivalent of Table 4, shown in Table 5. What we see are near identical values, though there is a slight difference. The minimums never reach the same lows, and the maximums never reach the same highs. However, the values throughout the 1st and 3rd quartiles are consistently above those from the census.

Metric	Total Units	Vacant Units	Vacancy Rate
Minimum	2011	296	0.0500
1st Quartile	4934	610.5	0.0900
Median	7199	776	0.0107
Mean	14220	1352.3	0.1210
3rd Quartile	11048	1370	0.1422
Maximum	204110	12985	0.3964

Table 5: Summary of the housing data estimates from the ACS. (N=99)

Examining the summary of the errors in Table 6 shows some extremely interesting behavior. The standard errors of the vacancy estimates are far higher than the total unit estimates. On average, the ratio of the vacant standard error to the total unit standard error is around 2.8, and goes all the way up to 7.1. This is before even accounting for the fact that the vacant unit estimates are far smaller, being a subset of the total. What this means is the vacancy unit estimates are far less precise than the total estimates are, and this lack of precision carries to the proportion as well. We can see that the relative standard errors for the vacancy estimates are nearly identical to those of the vacancy rates.

Metric	Total SE	Vacant SE	Rate SE	Relative Total SE	Relative Vacant SE	Relative Rate SE
Minimum	14.59	43.77	0.003	0.000959	0.02784	0.02782
Q1	27.36	79.64	0.011	0.003855	0.10164	0.10164
Median	42.55	103.34	0.011	0.005198	0.12230	0.12230
Mean	50.85	129.06	0.014	0.005782	0.12231	0.12231
Q3	60.49	142.55	0.017	0.006975	0.14533	0.14532
Maximum	265.65	618.24	0.027	0.024768	0.25127	0.25127

Table 6: Summary of the estimate standard errors and relative standard errors from the ACS. The relative columns are calculated using (SE / Estimate) for each given estimate. Some adjustments to metric and column names made to preserve space.

Examining the relationships between the estimates, their corresponding standard errors and relative standard errors in Figure 4 shows us a few things. Firstly, it shows that the magnitude of standard errors tend to rise alongside the magnitude of the estimates themselves, and that this is consistent across all 3 types of estimates. Second, it shows that the opposite relationship arises when the standard errors are instead looked at relative to the size of the estimates themselves. In other words, larger estimates tend to be more precise. The counties with less housing units have larger errors in proportion to their estimates. This intutively makes sense as the samples themselves are smaller, there is less data to go off of for the estimates. It is worth noting that this does not mean that the larger estimates are more accurate, as no comparison has been made yet to the true values from the census.

The takeaway from these findings is that the smaller counties are going to be worth looking at closely in the model comparisons. In theory, these are the counties with the most

room for growth as they can pull from the higher precision in other counties with more housing units.

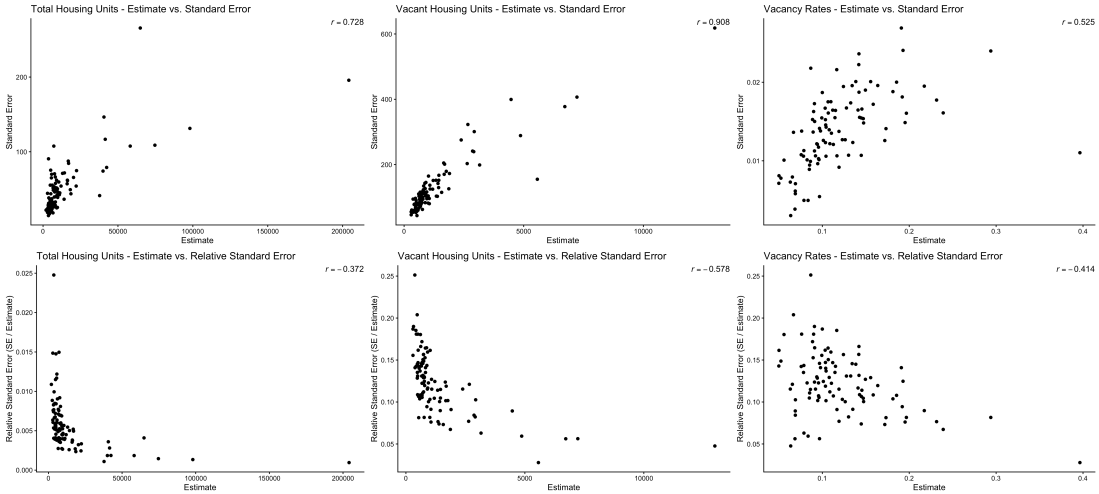


Figure 4: Plots showing the relationship estimates share with their standard errors and relative standard errors (SE / Estimate).

The distributions of the response and errors can be seen in Figure 5. The untransformed response is unimodal with a long right tail with a small bump in density around 0.4. The transformed response is very similar in distribution, but with a slightly less extreme right tail. Overall, it looks slightly more normally distributed than the untransformed response.

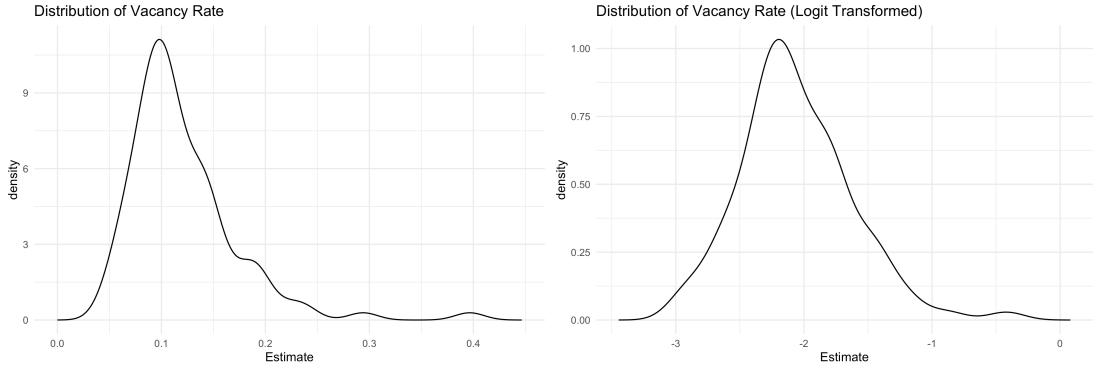


Figure 5: Distribution of the response before and after the logit transformation.

As for the errors shown in Figure 6, the shape stays mostly the same both before and after the response is transformed. The errors of the logit transformed response do show a couple additional bumps that aren't present in the original errors, though that may be more of an artifact of the plotting methodology than a meaningful change in distribution.

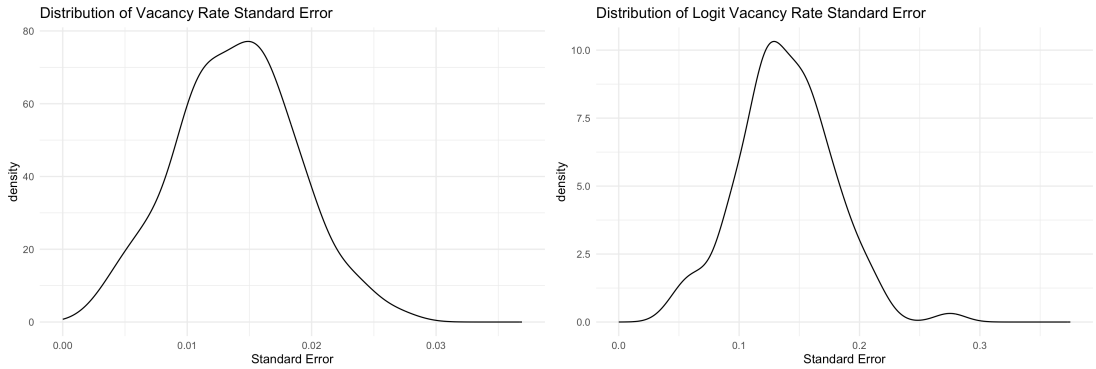


Figure 6: Distribution of the standard errors before and after the logit transformation of the response.

2.3.4. Tests of Spatial Autocorrelation for Housing Vacancy Rates

To assess the utility of an ICAR component in the Fay-Herriot model, Moran's I tests were performed on both the ACS response and the Census response separately to test for spatial autocorrelation. For this, the `moran.test` function was used from the `spdep` package on CRAN. For each dataset, two tests were performed under the constant risk hypothesis. One test uses binary neighbor encoding, another uses row standardization. The hypotheses for these tests are as follows:

$$H_0 : \rho = 0$$

$$H_a : \rho > 0$$

From the results in the figure below, both forms of encoding from both datasets reach the same conclusions at the $\alpha = 0.05$ level. There does seem to be evidence of positive autocorrelation of housing vacancy rates under the constant risk hypothesis.

Data	Matrix Encoding	Moran's I	Expected I	Variance	Z-Score	p-value
ACS	Binary	0.147	−0.01	0.0034	2.79	0.0026
ACS	Row Standardized	0.160	−0.01	0.0034	2.94	0.0016
Census	Binary	0.130	−0.01	0.0034	2.48	0.0066
Census	Row Standardized	0.142	−0.01	0.0034	2.62	0.0043

Table 7: Moran's I results from both the ACS and Census datasets. These tests performed under the normality assumption and constant risk hypothesis.

It is worth noting that these tests were primarily used as a diagnostic and a quick check of method feasibility. The risk from incorrectly rejecting the null hypothesis is low, and so additional tests were not performed. What these results show is that an ICAR component in the Fay-Herriot model does have some grounding, it isn't being applied to a response with no evidence of spatial autocorrelation.

2.3.5. NASS Crop Data Layer

To help motivate many of the decisions made in Section 2.2.2, analysis will begin with the crop data layer itself. Examining Table 8 shows that the makeup of land in Iowa is very different from that of the USA as a whole. Corn and Soybeans make up 38% and 27% of

the states land coverage respectively. Adding those up, approximately 65% of Iowas land coverage is corn or soybeans. Neither of these even reach 5% of the land coverage in the USA overall. The plots shown in Figure 7 visualize the similar spatial distributions and land coverage of corn and soybeans across the state. Their similarities and high prevelance, alongside their importance to Iowas overall agriculture gave the motivation for using them as a proxy for agricultural coverage per county.

USA		**Iowa**	
Crop	**Percentage**	**Crop**	**Percentage**
Shrubland	23.88	Corn	38.29
Grassland/Pasture	16.24	Soybeans	26.52
Evergreen Forest	12.58	Grassland/Pasture	13.73
Deciduous Forest	10.93	Deciduous Forest	8.43
Corn	4.85	Developed/Open Space	3.52
Woody Wetlands	4.59	Developed/Low Intensity	1.79
Soybeans	4.39	Herbaceous Wetlands	1.34

Table 8: Top crops by percentage of land classification for both the full USA CDL and Iowa subset. Top 7 rows.

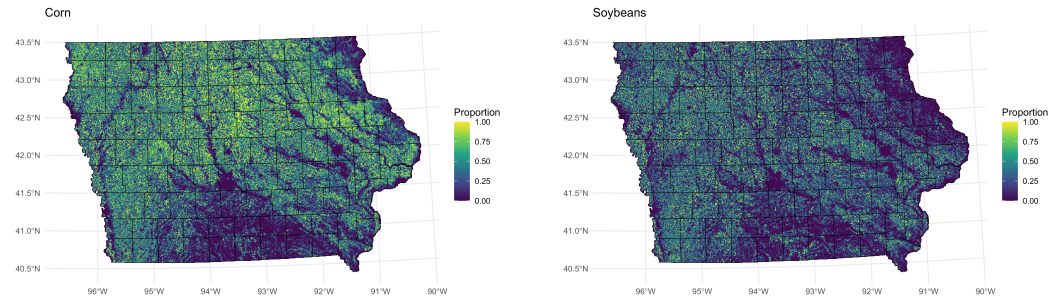


Figure 7: Plots showcasing the aggregated spatial distribution of corn and soybeans. Aggregation performed on each crop separately taking the mean of every 25x25 pixel area.

Amongst the various CDL classifications, there are ones that are referred to as "double crops". These are classifications with two different crops in them. There are 6 of these classifications total, examples including values like corn/soybeans and soybeans/oats. These were considered for inclusion in the covariate, though upon further inspection they were far too infrequent to justify. All 6 values total up to only around $4,000$ pixels total across all of Iowa. The max percentage for a double crop in a county was 0.042%. As such, these were ignored in all later work.

The top crops were examined for all counties to check for any other crops worth consideration. The top 3 crops for all 99 counties simplify down to only 6 cdl classifications. Those being corn, soybeans, developed/low intensity, developed/open space, grassland/pasture and deciduous forest. Nothing here appeared to be worth consideration for inclusion, so the decision was made to stick exclusively to corn and soybeans.

Narrowing down to the key crops at the county level, the overview in Table 9 shows that even the minimums show a large percentage of land coverage for the two key crops. At the most extreme almost 90% of land is classified as either corn or soybeans.

Statistic	Corn (%)	Soybeans (%)	Corn or Soybeans (%)
Minimum	8.58	6.53	19.95
1st Quartile	29.53	21.14	52.51
Median	41.32	27.47	69.36
Mean	37.73	26.42	64.15
3rd Quartile	47.73	32.22	77.84
Maximum	57.5	38.46	88.51

Table 9: Summary of the crop distribution across all Iowa counties.

Visualizing this information in Figure 8 shows some indication of spatial autocorrelation of land coverage. Many of the highest coverage counties are grouped together near the north west part of the state, and the lowest coverage counties are grouped in the south.

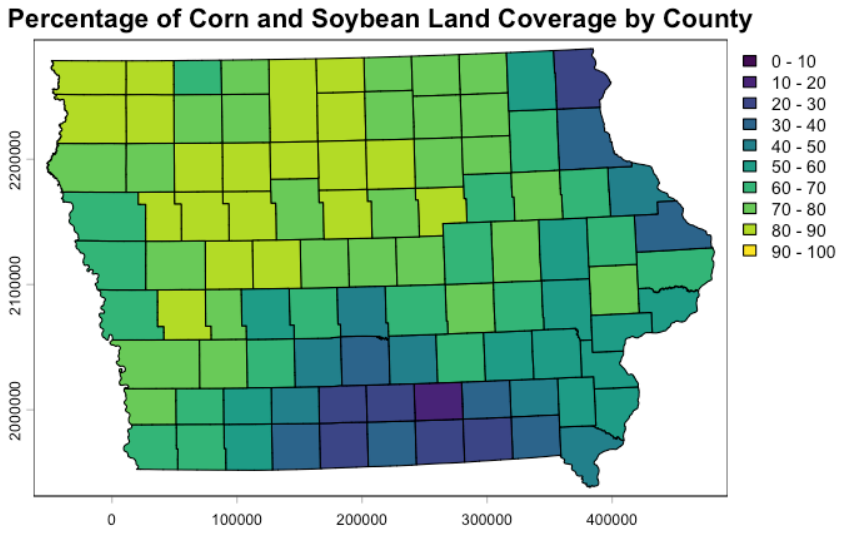


Figure 8: Plot of land coverage per county.

3. Results

4. Conclusion

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